

Book Review

J. Haugeland, *Artificial Intelligence: The Very Idea* (MIT Press, Cambridge, MA, 1985); 287 pp.

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John Haugeland's new book is a splendid layperson's introduction to the foundational problems of artificial intelligence. While its scope is very broad, ranging from an overview of the history of Western philosophy to an analysis of Turing's thesis, it nevertheless has a unifying theme: to assess the idea that underpins the enterprise of AI—the proposition that *Thinking is computing*.

True to a philosopher's professional ethics, Haugeland presents his analysis of this foundational idea as impartially as he can. He vows not to take sides on whether artificial intelligence is bound to succeed or doomed to failure. On the contrary, his central thesis, first stated in the introduction and reiterated throughout the book, is that the foundational proposition of AI is *empirical*. The possibility of artificial intelligence, the possibility that "machines can replicate human thinking," is neither a priori absurd nor a priori obvious: it is a matter of experience. Moreover, there is at present insufficient evidence to balance the scale one way or the other.

Haugeland does well to frame the AI enterprise in its broad philosophical context, a task too often ignored in the AI literature. He begins in Chapter 1 with an overview of the history of Western philosophy as it pertains to the idea that "thinking (intellection) essentially is rational manipulation of mental symbols (viz., ideas)" (p. 4). Thomas Hobbes, in this account, is singled out as the grandfather of AI. In Hobbes' philosophy, everything, including human reasoning, could be explained in terms of matter in motion. Indeed Hobbes had hoped to do for the science of the mind—reduce it to the science of motion—what Galileo had done for the science of motion—reduce it to the science of geometry. The cost of this enterprise, according to Haugeland, was having to give up on a theory of how thoughts can be meaningful: "he cannot tell" Haugeland writes "between minds and books" (p. 25). If human beings are merely pieces of matter in motion then indeed there is no essential distinction between a mental thought and the written word. But, Haugeland suggests, there is a distinction which Hobbesian materialism fails to draw,

Artificial Intelligence 29 (1986) 349–353

namely that thoughts *have* meanings whereas books *express* meanings. The materialism underlying AI science, on the other hand, appears to account for the concept of meaning to Haugeland's satisfaction, though the reasons for this are never terribly clear.

Any account of the rationalist heritage of AI would be incomplete without reference to (and reverence for) Descartes. Haugeland applauds the French philosopher's discovery of the distinction between the symbol and what it symbolizes—a discovery owed to the observation that algebraic relations among physical properties are expressible in purely geometric terms. In the same vein, thoughts are, to use Haugeland's anachronistic phrase, "symbolic representations" of objects and their relations (be they physical or mathematical). Seen in that light Descartes was responsible for discovering the foundational concepts of AI.

But Descartes was also influential for inventing the mind/body problem. If, as he believed, the mind is distinct from the body (indeed, for him they were different "substances"), an explanation is required as to how they interact—that when I will my arm to move, it moves. AI's answer to Descartes' dualist philosophy, says Haugeland, is materialism: the theory that all phenomena (including mental phenomena) are "made up" of matter. It is worth noting that Haugeland is not taking sides on the materialism/dualism debate. He simply makes the case that AI makes sense of the idea that the mind is a machine.

There is, however a difficulty with the naively materialist (or Hobbesian) reply to Descartes that Haugeland refers to as "the Paradox of Mechanical Reason"—the problem of how it is possible for a mechanical object to understand *meanings*. Mechanized reasoning, as Haugeland explains in detail in Chapters 2 and 3, is the manipulation of meaningful symbols using a system of rules. Now if the rule manipulator (i.e. the machine) attaches no meaning to the symbols it manipulates then we can't really say it is reasoning. On the other hand if it performed its manipulations according to the meaning of the rules and symbols it manipulates, it wouldn't be mechanical since "meanings (whatever exactly they are) don't exert physical forces" (p. 39). Moreover, even if meanings are conferred to rules by the machine, the question arises about how those meanings are themselves "understood" by the machine. We would have to imagine "homunculi" inside the machine that understand the meanings of the rules, thereby leading to an infinite regress of circular explanations.

Although the "Paradox of Mechanical Reason" appears to be something of a straw man, Haugeland's demonstration of how the paradigm of AI can resolve it merits some attention. In particular, he uses the thesis that "a computer is an *interpreted formal system*" to explain how "meanings" can be embodied in a machine and how reasoning can be mechanized. A formal system is loosely defined as a system of rules that define the possible configurations of the tokens (or "individuals") in the formal system. Games, like chess and tic-tac-toe, for example, are formal systems that provide rules for manipulating chess pieces or "○" and "×" symbols respectively.

There are three essential features of formal games: that they “manipulate tokens” (pieces, symbols), that they are “digital” and that they are “finitely playable.” Games that manipulate tokens are commonplace and Haugeland’s explanation of token-manipulating rules is straightforward: usually the purpose of a formal game is to arrive at a recognizable configuration of tokens (checkmate, three “○” or “×”s in a row, etc). The “digital” character of formal games is less intuitive. A digital operation is usually thought of as choice function on the discrete states of a system. But Haugeland generalizes the notion of a digital operation to be “a set of positive and reliable techniques for producing (‘writing’) and reidentifying (‘reading’) tokens or configurations of tokens from a prespecified set of [token] types.” (p. 53) This implies, for example, that a Shakespeare sonnet is a digital system because its tokens are alphabetic characters that can be recognized (read) or reproduced (written) in a positive (unambiguous) and reliable way. A Rembrandt painting, on the other hand, does not have “digital” properties because of its colors and textures. Haugeland’s reasoning is a bit weak here for he does not explain why a painting cannot be “digitized,” even in principle. This is particularly puzzling in view of the flexibility of his definition of a “positive” technique for recognizing tokens. Presumably, the ability to reflect rays of light of exactly the right frequency at precisely the right intensities should count as a “positive” technique for identifying graphics patterns.

The concept of “finite playability” and its relation to algorithms is clearly explained while avoiding all technicalities. A formal game is finitely playable if any player can determine, in a finite number of steps, whether a given move is legal and if the player can produce a legal move when such exists. This discussion leads to a natural definition of the essence of computer programs—the *algorithm*. An algorithm is simply a recipe for playing a finitely playable formal game. This is followed by a discussion of *schedules*, *conditional branching*, *nondeterministic algorithms*, and *heuristics*, all of which are explained with great ease and intuitive examples.

Haugeland complements this account of automatic formal systems with an entertaining chapter on the *semantics* of formal systems, thus returning once again to the problem of meaning. He draws the usual distinctions between syntax and semantics, successfully unravels the concept of “interpretation” as it applies to the axioms of arithmetic and logic, and all the while avoids the use of any mathematical or logical symbolism. It is also in this chapter that we are shown how “GOFAI” (good old fashioned artificial intelligence—that is, the AI enterprise conceived in the Hobbesian spirit) succeeds in solving the “Paradox of Mechanical Reason” mentioned earlier. The paradox arose, says Haugeland, because we were not looking at the problematic nesting of rule manipulators (leading to infinite regress) from the right point of view. Regarded as purely syntactic devices, the problem of how each “homunculus” responsible for rule manipulations understands the “meaning” of rules simply disappears. Each rule manipulator can be thought of as being made up of

subrule manipulators, each of which has a subsemantics until eventually the only rule manipulations are “mechanical” (adding binary digits).

While this account of “meaning” is appealing, there is little evidence to indicate that it can be extended to intentional objects generally. He does not make the case, for example, that desires or beliefs can be treated in the same way as rule governed linguistic behavior. One gets the distinct impression that the problem of meaning has been dismissed rather than solved.

Chapter 4, Computer architecture, is historically informative. Haugeland covers some familiar material starting with Babbage’s analytical engine, and continuing with Turing machines and universal Turing machines, von Neumann machines, LISP machines, and production systems. Conspicuously absent from the discussion, however, are parallel architectures (vector processors, multiple processors systems, hypercubes, and the like). Haugeland’s apologies in the acknowledgements do not quite make up for this lacuna, particularly since he explicitly observes in this chapter that one of the essential differences between a von Neumann computer and the brain is the parallelism inherent in cerebral neuron connections.

Chapter 5, Real machines, could have been entitled, What machines can’t do. Here, Haugeland exposes the limitations inherent in some of the approaches that have been taken in the history of AI research, all the while reminding the reader that the failure of these approaches do not condemn the project of AI (or even refute the fundamental proposition that thinking is computing). The discovery that machine translation, cybernetics, automatic theorem proving (Newell’s general problem solver), micro-worlds (Winograd’s SHRDLU), and expert systems all fail to deliver the promise of AI is not a sufficient reason to damn the project as unrealistic or even impossible. If anything, it has shown, according to Haugeland, that specialized systems implementing brute force methods have limited applicability. *Real* GOFAI, therefore, needs to synthesize the intelligence in Real people, the topic of the last chapter.

The discussion of Real people is not intended to be an indictment of GOFAI: on the contrary, it points to many areas that need to be sorted out before Real AI is possible. Real people, he points out, understand *conversational implicatures* (context-dependent presuppositions), they can disambiguate according to context, they can manipulate mental images in their minds, and they have a sense of personal identity. Of these capabilities one may well ask which is the most important for an AI system. The answer is not definite, but Haugeland thinks that personal identity or “ego involvement” ranks among the most important. “Ego involvement” he writes “may be an integral part of understanding.” Having a sense of humor, he suggests, requires some kind of ego involvement, or at least the capacity to imagine what it is like to be in someone else’s shoes. At any rate, there are global aspects of intelligent behavior that require the replication of human capacities and that GOFAI hasn’t yet succeeded in even mimicing.

The main thesis of the book is undeniably sound. The very idea of AI may or may not be successful: we can only find out by trying. Haugeland's explanation of why this is so is usually informative, even if sometimes cursory: there are many details which beg for further analysis. But the net result is convincing: the case is clearly made that AI is possible but it is equally clear that the proof of the pudding is in the eating.